

Association of serum Wnt1-inducible signaling pathway protein 1 (WISP1) levels with osteoarthritis and obesity-related inflammation in obese adults

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Serum Wnt1-Inducible Signaling Pathway Protein 1 (WISP1) Levels and Osteoarthritis in Obese Adults: Insight into the Inflammatory Link Between Obesity and Osteoarthritis Association of Serum Wnt1-Inducible Signaling Pathway Protein 1 (WISP1) Levels with Osteoarthritis and Obesity-Related Inflammation in Obese Adults

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Short Title: Serum WISP1 and Osteoarthritis in Obesity

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Abstract

Background: Osteoarthritis (OA) is increasingly recognized as a metabolically influenced inflammatory disease, particularly in the context of obesity. Wnt1-inducible signaling pathway protein 1 (WISP1), a downstream mediator of Wnt/ β -catenin signaling, has been implicated in adipose tissue inflammation and cartilage remodeling; however, clinical data regarding circulating serum WISP1 levels in obesity-associated OA remain limited.

This study aimed to investigate the relationship between serum WISP1 concentrations and OA status in obese adults. **Methods:** This cross-sectional study included 180 participants classified into three groups: healthy non-obese controls (n = 60), obese individuals without OA (n = 60), and obese individuals with OA (n = 60). Clinical, anthropometric, metabolic, and inflammatory parameters were recorded. Serum WISP1 concentrations were measured using enzyme-linked immunosorbent assay (ELISA). Group comparisons, correlation analyses, logistic regression models, and receiver operating characteristic (ROC) analyses were performed. **Results:** Serum WISP1 concentrations differed significantly among groups, showing a stepwise increase from healthy controls to obese OA (-) and obese OA (+) participants ($p < 0.001$). WISP1 remained independently associated with osteoarthritis among obese individuals after adjustment for age, sex, body mass index, HOMA-IR, diuretic use and C-reactive protein (OR = 1.003; 95% CI: 1.001-1.005). Although obese participants demonstrated adverse metabolic and inflammatory profiles compared with controls, most metabolic parameters did not distinguish OA subgroups within obesity. ROC analysis demonstrated ~~strong~~ good discriminatory performance of WISP1 for distinguishing obese OA (+) from ~~versus healthy controls~~ (AUC = ~~0.980~~) and ~~moderate discrimination between obese OA (+) and obese OA (-)~~ groups (AUC = 0.807, 95% CI: 0.727-0.879, $p < 0.001$). **Conclusions:** Serum WISP1 concentrations are elevated in obese individuals with osteoarthritis and are independently associated with OA status beyond

conventional measures of adiposity and systemic inflammation. These findings support a potential role of suggest that WISP1 may serve as a potential biomarker associated with metabolic-inflammatory processes reflecting metabolic-inflammatory processes in obesity-associated osteoarthritis. Given the cross-sectional nature of the study, the observed findings should be interpreted as associations, and causal relationships cannot be inferred. Further longitudinal and externally validated studies are needed to clarify its clinical utility and biological significance. **Keywords:** osteoarthritis; obesity; WISP1; adipokine; inflammation.

Introduction

Osteoarthritis (OA) is the most prevalent degenerative joint disorder worldwide and represents a leading cause of chronic pain, disability, and reduced quality of life among aging populations (1). Pathologically, OA is characterized by progressive articular cartilage loss, subchondral bone remodeling, osteophyte formation, and varying degrees of synovial inflammation, ultimately resulting in impaired joint function (2). In parallel with global demographic aging and the rising prevalence of obesity, the overall burden of OA is projected to increase substantially in the coming decades (3). Although historically regarded as a consequence of mechanical “wear and tear,” OA is now widely recognized as a complex, multifactorial disease in which biomechanical, metabolic, and inflammatory factors interact to drive structural joint deterioration (4).

Obesity is among the most influential modifiable risk factors for OA. Beyond increasing mechanical loading on weight-bearing joints, adipose tissue functions as a metabolically active endocrine organ that secretes adipokines and pro-inflammatory mediators capable of influencing cartilage and synovial biology (5). Elevated circulating levels of cytokines such as interleukin-6 (IL-6) and tumor necrosis factor-alpha (TNF- α) have been associated with synovial inflammation and cartilage degradation in OA (6). Moreover, increased expression of leptin and its functional receptor (Ob-Rb) has been demonstrated in osteoarthritic cartilage, suggesting a role for adipokine-mediated modulation of matrix turnover (7). These observations have contributed to the concept of “metabolic OA,” wherein obesity-related low-grade systemic inflammation synergizes with mechanical stress to exacerbate structural joint damage (8).

Recent investigations have increasingly emphasized the inflammatory component of OA, suggesting that its role extends beyond a secondary response to structural damage. Synovial tissue frequently exhibits macrophage and T-lymphocyte activation, and synovitis has been associated with structural progression and pain severity (9). Pro-inflammatory cytokines and chemokines stimulate the production of catabolic enzymes, particularly matrix metalloproteinases (MMPs) and aggrecanases (notably ADAMTS-4 and ADAMTS-5), thereby accelerating degradation of type II collagen and aggrecan (10). These processes contribute to a self-perpetuating cycle of matrix breakdown, cellular stress, and impaired repair capacity (11).

Within this inflammatory-metabolic framework, the Wnt/ β -catenin signaling pathway has emerged as a key regulator of cartilage homeostasis. Physiological Wnt activity is essential for cartilage development; however, dysregulated activation promotes chondrocyte hypertrophy, catabolic enzyme expression and osteophyte formation in OA (12). Experimental studies demonstrate that excessive Wnt signaling accelerates cartilage degeneration, whereas its inhibition mitigates structural damage, underscoring its relevance in OA progression (13).

Wnt1-inducible signaling pathway protein 1 (WISP1) is a matricellular protein and downstream effector of Wnt/ β -catenin signaling. As CCN4, a member of the CCN family of matricellular proteins, WISP1 interacts with extracellular matrix components and cell-surface receptors to regulate cellular proliferation, fibrosis, and tissue remodeling (14). In human osteoarthritic cartilage, WISP1 expression is increased in lesioned regions compared with preserved areas and is associated with disease-related alterations in DNA methylation. In vitro studies indicate that WISP1 exposure in chondrocytes reduces extracellular matrix synthesis while enhancing expression of matrix-degrading enzymes such as MMPs and ADAMTS (15). Furthermore, Wisp1-deficient mice develop attenuated cartilage degeneration following experimental OA induction, supporting its involvement in structural joint pathology (16).

Beyond its role in joint tissues, WISP1 has also been characterized as an adipokine linked to obesity and metabolic dysfunction. Increased WISP1

expression in adipose tissue of obese individuals correlates with markers of inflammation and insulin resistance (17). Serum WISP1 concentrations have been associated with body mass index (BMI) and metabolic parameters in human cohorts (17-20), and experimental studies demonstrate that WISP1 can interfere with insulin signaling pathways and modulate inflammatory cascades (18). These findings position WISP1 at the intersection of adiposity, insulin resistance, and low-grade systemic inflammation.

Despite accumulating experimental and tissue-level evidence implicating WISP1 in cartilage remodeling and metabolic inflammation, clinical data regarding serum WISP1 concentrations in OA remain scarce. Most available studies have focused on intra-articular expression or experimental models, whereas the relationship between systemic WISP1 levels and OA status in well-characterized obese populations has not been adequately examined. It therefore remains unclear whether elevated serum WISP1 reflects adiposity-related metabolic activation alone or is also associated with OA at the systemic level.

Therefore, the present study aimed to evaluate serum WISP1 concentrations in obese individuals with and without OA and to explore their association with clinical and inflammatory parameters. By addressing the gap between experimental findings and serum biomarker data, this study seeks to clarify the potential relevance of WISP1 within the metabolic-inflammatory framework of obesity-associated osteoarthritis.

Materials and Methods

Study Design

This single-center, cross-sectional observational study was conducted at the Department of Internal Medicine, Kanuni Sultan Suleyman Training and Research Hospital. The study protocol was reviewed and approved by the Institutional Clinical Research Ethics Committee (approval date: 20 February 2025; decision no: E-80929729-000-269138296) and was performed in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants prior to enrollment. An a priori power analysis was conducted using G*Power software (version 3.1.9.2; Heinrich Heine University, Düsseldorf, Germany). Based on variance estimates derived from a pilot subset and assuming a one-way ANOVA (fixed effects, omnibus), the effect size was calculated as Cohen's $f = 0.31$. With a significance level (α) of 0.05, statistical power ($1 - \beta$) set at 0.95, and three study groups, the minimum required total sample size was 164 participants. A total of 180 individuals were ultimately included, exceeding the calculated requirement.

Study Population

A total of 180 participants aged 18–80 years were consecutively recruited during the study period. Participants were categorized into three predefined groups ($n = 60$ per group) according to BMI and OA status: Group 1, healthy non-obese controls (HC); Group 2, obese individuals without OA [obese OA (-)]; and Group 3, obese individuals with OA [obese OA (+)]. Groups were frequency-balanced with respect to age and sex distribution. Obesity was defined according to World Health Organization criteria as BMI ≥ 30 .

kg/m² (21). Anthropometric measurements were obtained under standardized conditions using calibrated equipment. The diagnosis of OA was established based on compatible clinical symptoms and radiographic findings, including joint space narrowing, osteophyte formation, and subchondral sclerosis, in accordance with contemporary American College of Rheumatology clinical classification frameworks for symptomatic OA (22). Radiographic severity was graded using the Kellgren-Lawrence classification, in and only patients with knee definite radiographic OA (grade ≥ 2), where a grade ≥ 2 was considered indicative of definite radiographic OA (23), although the overall diagnosis was based on combined clinical and radiographic assessment were included in the OA group (23). All participants presenting with musculoskeletal complaints were evaluated by an experienced orthopedic specialist. Radiographs were assessed by the same orthopedist, and in cases of diagnostic uncertainty, images were independently reviewed by a second orthopedist to reach consensus. Laboratory personnel were blinded to clinical group allocation.

Exclusion Criteria

Exclusion criteria included secondary arthritis (rheumatoid arthritis, gout, psoriatic arthritis, post-traumatic arthritis), prior major joint surgery, congenital musculoskeletal abnormalities, severe systemic disease (advanced hepatic, renal, or cardiac failure), active malignancy, diagnosed diabetes mellitus, chronic inflammatory or autoimmune disorders, long-term systemic corticosteroid or immunosuppressive therapy, untreated endocrine

disorders, active infection, pregnancy or lactation, and inability to provide informed consent.

Data Collection

Demographic data (age, sex, family history), comorbidities, and medication use were recorded from electronic medical records. Body weight and height were measured using standardized calibrated devices, and BMI was calculated as weight (kg) divided by height squared (m^2). Body composition analysis was performed using a multi-frequency bioelectrical impedance analyzer (Tanita MC-780MA, Tokyo, Japan), providing measurements of total body fat percentage, visceral fat index, fat-free mass, and skeletal muscle mass. All measurements were conducted in the morning following an overnight fast of at least 8 hours. Participants were instructed to avoid caffeine and strenuous physical activity for 24 hours prior to assessment. Measurements were performed by the same trained operator according to manufacturer guidelines. Blood pressure was measured in the seated position after a minimum of 10 minutes of rest. Two readings were obtained and averaged for analysis.

Sample Collection

Fasting venous blood samples were collected between 08:00 and 09:00 a.m. after an overnight fast of 8–10 hours. Samples were drawn into plain tubes, centrifuged at 4°C (relative centrifugal force equivalent to approximately $3000 \times g$) for 10 minutes, and serum aliquots were stored at

–80°C until analysis. All samples were thawed once and analyzed in batch during a single analytical session to minimize inter-assay variability. Laboratory staff were blinded to group allocation. Routine biochemical parameters including fasting plasma glucose, serum creatinine, alanine aminotransferase (ALT), albumin, total cholesterol, triglycerides, high-density lipoprotein-cholesterol (HDLc), and low-density lipoprotein-cholesterol (LDLc) were measured using enzymatic colorimetric methods on an automated analyzer (Roche Cobas 8000 c702; Roche Diagnostics, Mannheim, Germany). High-sensitivity C-reactive protein (hsCRP) was measured using an immunoturbidimetric assay on the same platform.

Measurement of Serum WISP1 Levels

Serum WISP1 concentrations were measured as the primary biomarker using a double-antibody sandwich enzyme-linked immunosorbent assay (ELISA) kit (Human WISP1 ELISA Kit, Sunred Biological Technology Co., Shanghai, China; Catalog No. 201-12-2360) according to the manufacturer's instructions. Samples and standards were analyzed in duplicate. Optical density was measured at 450 nm using a calibrated microplate reader (BioTek ELx808, Winooski, VT, USA), and mean values were used for statistical analysis. The assay sensitivity was 20.2 pg/mL, with a measurable range of 22–6000 pg/mL. Serum WISP1 concentrations were expressed in pg/mL. The intra-assay and inter-assay coefficients of variation were <10% and <12%, respectively. Repeated freeze-thaw cycles were avoided to preserve analyte stability. All samples were analyzed in duplicate within a

single analytical batch during the same experimental session to minimize variability.

Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics version 26.0 (IBM Corp., Armonk, NY, USA). Normality of continuous variables was assessed using the Kolmogorov-Smirnov test. Normally distributed variables were expressed as mean $\pm$ standard deviation (SD), and non-normally distributed variables as median (interquartile range, Q1-Q3). Categorical variables were reported as frequencies and percentages. Group comparisons were performed using one-way ANOVA for normally distributed variables and the Kruskal-Wallis test for non-normally distributed variables. Post hoc comparisons were conducted using Bonferroni correction (ANOVA) or Mann-Whitney U test (Kruskal-Wallis). Categorical variables were analyzed using Pearson's chi-square test. The distribution of serum WISP1 levels across groups was visually presented using boxplots generated in Python (version 3.13, 64-bit) with seaborn and matplotlib libraries. Spearman's rank correlation analysis was used to assess associations between serum WISP1 and clinical/laboratory variables. Univariate and multivariable binary logistic regression analyses were conducted to identify factors independently associated with OA among obese participants. Univariate and multivariable binary logistic regression analyses were performed to identify factors independently associated with OA among obese participants. In the multivariable model, age, gender, BMI, CRP, HOMA-IR, diuretic use, and

serum WISP1 concentration were entered simultaneously based on their clinical and biological relevance. Model fit was assessed using the Omnibus test, Hosmer-Lemeshow goodness-of-fit test, and pseudo- R^2 statistics. In the binary logistic regression analysis, 60 outcome events were included, corresponding to an events per variable ratio of approximately 8.6. ~~Variables significant in univariate analysis were entered simultaneously into multivariable models. Model fit was evaluated using the Omnibus test and pseudo- R^2 statistics.~~ Multicollinearity was assessed using tolerance and variance inflation factor (VIF) values. Adjusted odds ratios (ORs) with 95% confidence intervals (CIs) were reported. Multinomial logistic regression analysis was performed to determine independent variables differentiating HC, obese OA (-), and obese OA (+), with HC as the reference category. Receiver operating characteristic (ROC) curve analysis was used to evaluate the discriminatory performance of serum WISP1. The area under the curve (AUC) with 95% CIs was calculated, and the optimal cut-off value was determined using the Youden index. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were reported. A p value <0.05 was considered statistically significant.

Results

A total of 180 individuals were included in the study, comprising 60 healthy non-obese controls, 60 obese individuals without OA, and 60 obese individuals with OA. The median age of the study population was 54 years. Of the participants, 152 (84.4%) were female and 28 (15.6%) were male. Within

the OA group, 44 patients had knee OA, and 16 had non-knee OA. Among patients with knee OA, 13 (29.5%) were classified as Kellgren-Lawrence grade 2, 24 (54.5%) as grade 3, and 7 (15.9%) as grade 4.

Significant differences were observed in anthropometric and body composition parameters. BMI, waist circumference, waist-to-hip ratio, muscle mass, visceral fat level, and fat-free mass index were significantly higher in both obese OA (-) and obese OA (+) groups compared with HC (all $p < 0.001$), with no significant difference between the two obese groups. Body fat percentage was significantly highest in the obese OA (+) group, lower in the obese OA (-) group, and lowest in HC ($p < 0.001$).

FPG, HOMA-IR, total cholesterol, LDLc, ALT, and triglyceride levels were higher in both obese groups than in healthy controls (all $p < 0.001$), whereas no significant differences were observed between obese OA (-) and obese OA (+). HDLc levels were lower in the obese OA (-) group than in controls ($p = 0.007$), with no significant difference between the two obese groups (Table 1).

Inflammatory markers showed a distinct pattern. CRP levels were lowest in controls, higher in obese OA (-), and highest in obese OA (+) ($p < 0.001$). Erythrocyte sedimentation rate was significantly higher in obese OA (+) compared with both obese OA (-) and HC ($p < 0.001$). White blood cell counts were significantly elevated in both obese groups compared with HC ($p < 0.001$), with no significant difference between obese OA (-) and obese OA (+). The detailed results are summarized in Table 1.

Serum WISP1 concentrations differed significantly among the study groups ($p < 0.001$). Median WISP1 levels were lowest in healthy controls (258.5 pg/mL), significantly higher in obese OA (-) participants (385.0 pg/mL), and highest in obese OA (+) individuals (525 pg/mL). Pairwise comparisons demonstrated that WISP1 concentrations were significantly elevated in both obese groups compared with controls, and were further significantly increased in obese OA (+) compared with obese OA (-) (all $p < 0.001$). The distribution of serum WISP1 concentrations across the three groups is illustrated in Figure 1.

When obese OA (+) participants were stratified according to OA localization, serum WISP1 concentrations remained significantly different across groups ($p < 0.001$). Median WISP1 levels were lowest in healthy controls (258.5 pg/mL), and significantly higher in both obese patients with knee OA (506 pg/mL) and those with non-knee OA (554 pg/mL). Pairwise comparisons demonstrated that both OA subgroups exhibited significantly elevated WISP1 concentrations compared with controls (all $p < 0.001$), whereas no significant difference was observed between knee and non-knee OA. Detailed subgroup data are presented in Table 2.

Spearman correlation analysis revealed a significant positive association between serum WISP1 concentrations and total cholesterol levels in healthy controls ($\rho = 0.285$, $p = 0.028$). In the obese OA (-) group, serum WISP1 concentrations were positively correlated with BMI ($\rho = 0.289$, $p = 0.025$). No other significant correlations were observed between serum

WISP1 concentrations and clinical, anthropometric, metabolic, or inflammatory parameters in any of the study groups.

In univariate logistic regression analysis, serum WISP1 concentration was significantly associated with the presence of OA among obese individuals ($p = 0.001$). Each 1 pg/mL ~~one-unit~~ increase in serum WISP1 concentration ~~level~~ was associated with a 1.003-fold increase in the odds of OA (OR = 1.003; 95% CI: 1.001-1.004). In the multivariable logistic regression model, the overall model was statistically significant ($\chi^2 = 32.572$ 39.000, $p < 0.001$). The explanatory power of the model was moderate, with a Cox & Snell R^2 of ~~0.238~~ 0.277 and a Nagelkerke R^2 of ~~0.317~~ 0.370. No evidence of multicollinearity was detected (all tolerance values > 0.20 and VIF < 5). After adjustment for age, sex, BMI, HOMA-IR, diuretic use and CRP, serum WISP1 concentration remained independently associated with OA ($p < 0.001$). Each 1 pg/mL ~~one-unit~~ increase in serum WISP1 was associated with a 1.003-fold increase in the odds of OA (OR = 1.003; 95% CI: 1.001-1.005) (Table 3). The detailed results of both the univariate and multivariable logistic regression analyses are presented in Table 3.

Multinomial logistic regression analysis was conducted using the healthy control group as the reference category. The overall model was statistically significant (likelihood ratio $\chi^2 = 228.255$, $p < 0.001$), with pseudo- R^2 values of 0.719 (Cox-Snell), 0.808 (Nagelkerke), and 0.577 (McFadden). Compared with healthy controls, serum WISP1 concentration was independently associated with both the obese OA (-) and obese OA (+)

groups (both $p < 0.001$ obese OA (-): OR = 1.025, 95% CI: 1.011-1.039, $p < 0.001$; obese OA (+): OR = 1.028, 95% CI: 1.014-1.042, $p < 0.001$). HOMA-IR and CRP levels were also independently associated with group classification in both comparisons (all $p < 0.05$ HOMA-IR: OR = 11.902, 95% CI: 1.775-79.801, $p = 0.011$; OR = 10.563, 95% CI: 1.575-70.837, $p = 0.015$; CRP: OR = 3.925, 95% CI: 1.642-9.382, $p = 0.002$; OR = 4.284, 95% CI: 1.788-10.263, $p = 0.001$), whereas age was not significantly associated. Detailed regression coefficients and 95% confidence intervals are presented in Table 4.

ROC curve analysis was performed to evaluate the discriminatory performance of serum WISP1 concentrations for identifying obese individuals with OA. ~~Separate ROC analyses were conducted for obese OA (+) versus obese OA (-) and for obese OA (+) versus HC. When obese OA (+) patients were compared with healthy controls, serum WISP1 concentrations demonstrated discriminatory performance, with an AUC of 0.980. A cut-off value of 401 pg/mL yielded 91.7% sensitivity and 93.3% specificity, with PPV of 93.2% and NPV of 91.8% (Figure 2).~~ As shown in Figure 3 2, serum WISP1 levels differentiated obese OA (+) patients from obese OA (-) individuals with an AUC of 0.807 (95% CI: 0.727-0.879, $p < 0.001$). The optimal cut-off value identified by the Youden index was 472 pg/mL, corresponding to 75.0% sensitivity and 81.7% specificity. The PPV and NPV were 80.4% and 76.6%, respectively.

Discussion

Serum WISP1 concentrations differed significantly across groups, being lowest in healthy controls, intermediate in obese individuals without OA, and highest in obese individuals with OA. This stepwise pattern is biologically plausible and consistent with previous literature. Murahovschi et al. (17) demonstrated increased WISP1 expression in adipose tissue in obesity, linking it to inflammatory activation and insulin resistance-related metabolic processes, while Mirr et al. (24) reported elevated circulating WISP1 levels associated with metabolic dysfunction. In the context of OA, Blom et al. (25) showed that Wnt pathway activation is associated with increased WISP1 expression and cartilage degeneration, and van den Bosch et al. (15) documented WISP1 overexpression in damaged osteoarthritic cartilage, including epigenetic regulation. Together, these findings provide a coherent framework in which the higher serum WISP1 levels observed in obese individuals with OA may be associated with ~~reflect~~ the coexistence ~~convergence~~ of metabolic activation and structural joint remodeling. Most prior evidence is derived from tissue-based or experimental studies, whereas the present analysis evaluates systemic serum concentrations. Therefore, serum WISP1 should not be interpreted as a direct surrogate of intra-articular expression but rather as a marker potentially associated with ~~reflecting~~ integrated metabolic and joint-related processes.

Subgroup analysis showed no significant difference in serum WISP1 between knee and non-knee OA, suggesting that the association is not confined to weight-bearing joints. Griffin et al. (26) emphasized that obesity-related OA

involves metabolic mechanisms beyond mechanical load, while Zhuo et al. (27) and Courties et al. (28) highlighted interactions between metabolic disturbances and joint tissues across multiple compartments. Similarly, Conde et al. (29) and Pottie et al. (30) demonstrated systemic effects of adipose-derived factors and complex metabolic-inflammatory pathways in OA. Within this context, the lack of localization-specific variation in WISP1 levels supports a systemic metabolic-inflammatory contribution across multiple joint compartments, although subgroup size and phenotypic heterogeneity warrant cautious interpretation.

Anthropometric findings support the interpretation of serum WISP1 levels. As expected, obese participants exhibited higher BMI and central adiposity compared with controls, consistent with established links between excess adiposity and OA risk (26-28). However, these measures did not clearly differentiate obese OA (+) from OA (-) individuals, suggesting that body mass alone may not determine OA status within obesity. This dissociation is consistent with prior work emphasizing metabolic and inflammatory alterations beyond mechanical load (29,30).

Compared with healthy controls, obese participants exhibited an adverse lipid profile characterized by higher LDLc and triglyceride levels and lower HDLc. However, lipid parameters did not significantly differ between obese OA (+) and obese OA (-) individuals, and triglyceride levels were numerically higher in the obese OA (-) group. This finding, in which triglyceride levels were higher in obese OA (-) individuals compared to those

with OA, may be explained by treatment-related factors, particularly the more frequent use of lipid-lowering therapies among patients with established OA due to greater healthcare utilization or comorbidity burden. In addition, differences in metabolic profiles or disease-related alterations not fully captured by conventional lipid parameters may also contribute to this observation. Therefore, this finding should be interpreted with caution.

Taken together with these observations, these findings suggest that dyslipidemia alone may not account for OA status within obesity. Zhuo et al. (27) and Courties et al. (28) identified lipid abnormalities as components of metabolic disturbances associated with OA. At the tissue level, Akagi et al. (31) demonstrated oxLDL and LOX-1 expression in articular cartilage, linking oxLDL to inflammatory signaling and catabolic activity, while Lee et al. (32) reported accelerated cartilage damage in experimental models. Within this framework, the lipid alterations observed in the present cohort are consistent with a metabolically altered environment in obesity; however, their limited discriminatory capacity between obese OA subgroups suggests that dyslipidemia may represent a background metabolic feature rather than a sole determinant of OA status.

Similarly, glycemic parameters were higher in obese participants but did not differentiate OA subgroups. Verzijl et al. (33) demonstrated that hyperglycemia promotes advanced glycation end-product accumulation in cartilage, while Schett et al. (34) and Khor et al. (35) reported associations between type 2 diabetes and OA outcomes, with heterogeneity and partial

mediation by obesity. These findings suggest that the glycemic alterations observed in the present cohort are consistent with a metabolically altered environment, although they do not independently define OA status.

Systemic inflammatory markers, including CRP and ESR, were elevated in obese OA (+), aligning with prior evidence linking low-grade inflammation to OA progression. Spector et al. (36) reported elevated CRP levels in early OA associated with disease progression, and Shadyab et al. (37) confirmed prospective associations between CRP and radiographic knee OA progression. Notably, serum WISP1 remained independently associated with OA after adjustment for CRP in multivariable models. Based on previous studies demonstrating the involvement of WISP1 in obesity-related inflammatory activation and cartilage remodeling pathways (15,17,25), this finding should be interpreted with caution ~~the persistence of its association after CRP adjustment suggests that WISP1 may reflect biological processes not fully represented by conventional acute-phase markers.~~

Correlation analyses showed limited and inconsistent associations between WISP1 and individual metabolic variables, suggesting that serum WISP1 cannot be explained by isolated indices such as adiposity, lipid parameters, or CRP. Rather than reflecting a single biochemical marker, WISP1 may represent more integrated biological processes, although subgroup heterogeneity and sample size may have limited detection of weaker associations.

Binary logistic regression demonstrated that serum WISP1 remained independently associated with OA among obese individuals after adjustment for covariates included in the multivariable model BMI and CRP. Because BMI reflects overall adiposity and CRP represents generalized systemic inflammation, the persistence of this association suggests that the relationship between WISP1 and osteoarthritis is not fully accounted for by mechanical load or classical acute-phase activity. This finding is consistent with prior mechanistic evidence (15,17,25). Within this framework, the independent association observed in the present model indicates supports the interpretation that serum WISP1 is associated with OA independently of age, gender, HOMA-IR, diuretic use, BMI and CRP within the limits of the model may reflect biological processes extending beyond conventional measures of adiposity and systemic inflammation. Although the odds ratio per 1 pg/mL increase in serum WISP1 appears numerically small, this reflects the scale of measurement. Importantly, when considered across a broader range, a 100 pg/mL increase corresponds to an approximately 35% increase in the odds of OA, suggesting that the cumulative effect of WISP1 may be clinically meaningful.

In the multivariable model, HOMA-IR and diuretic use were also independently associated with OA. The inverse association observed for HOMA-IR after adjustment, despite its non-significant effect in univariate analysis, may reflect interactions or redistribution of shared variance among metabolic covariates within the model rather than a direct protective

relationship, particularly given prior evidence linking WISP1-related pathways to insulin resistance (17). In contrast, the association between diuretic use and OA is likely to reflect underlying clinical characteristics or comorbidity patterns within the study population rather than a direct effect of the medication itself.

Multinomial logistic regression showed that higher serum WISP1 concentrations were independently associated with both obese groups compared with healthy controls, with stronger effects in obese OA (+). Although HOMA-IR and CRP showed larger effect sizes consistent with their roles in group differentiation, the association of WISP1 persisted after adjustment. ~~This finding suggests that the relationship between serum WISP1 and OA status is not fully accounted for by conventional indicators of metabolic dysfunction or low-grade inflammation.~~ Given that CRP is a non-specific acute-phase reactant, the persistence of this association indicates that it is not explained by CRP within the model. However, this finding should be interpreted strictly as a statistical association and does not imply distinct underlying biological mechanisms. These findings are consistent with studies showing that systemic metabolic disturbances, including insulin resistance and dyslipidemia, contribute to osteoarthritis beyond mechanical loading (27–28). Although WISP1 was not specifically examined, the present results align with this framework without implying causality. However, the relatively high pseudo- R^2 values observed in the multinomial model should be interpreted with caution, as they may reflect model-specific characteristics

and potential overfitting, particularly in the context of the sample size and number of predictors.

ROC analysis demonstrated discriminatory capacity of serum WISP1 concentrations for identifying obese individuals with OA. In comparisons between obese OA (+) patients and healthy controls, the AUC was 0.980, with a cut-off value of 401 pg/mL providing high sensitivity and specificity. When obese OA (+) patients were compared with obese OA (-) individuals, WISP1 demonstrated good discriminative performance discrimination was moderate (AUC = 0.807), supporting its ability to distinguish between these clinically relevant groups the Youden index identified an optimal cut-off of 472 pg/mL with balanced sensitivity and specificity. The Youden index identified an optimal cut-off value of 472 pg/mL, with a reasonable balance between sensitivity and specificity. As emphasized Bay-Jensen et al. (38), in their comprehensive review of soluble biomarkers in osteoarthritis, emphasized that discrimination observed in cross-sectional internal datasets should be interpreted cautiously, as it may reflect overlapping often reflects combined metabolic and inflammatory differences contrasts between groups rather than disease-specific predictive performance capacity. In the present study, separation between obese OA (+) and healthy controls likely captures broader differences in adiposity-related inflammation, whereas discrimination within the obese cohort may more directly reflect osteoarthritis-related biological variation. Accordingly Therefore, while the these ROC findings suggest that support potential utility of WISP1 may have

potential utility in phenotypic stratification within obese individuals, the clinical implications of this observation ~~they do not support diagnostic application they~~ should not be interpreted with caution as ~~evidence of diagnostic sufficiency~~. External validation, prospective studies ~~longitudinal outcome prediction~~, and comparison ~~against~~ with established biomarkers are warranted to further clarify its potential role ~~would be necessary before considering application~~.

Several limitations should be acknowledged. The cross-sectional design of the present study does not allow causal inference and limits ~~precludes~~ temporal ~~inference~~ interpretation; therefore, it cannot be determined whether elevated serum WISP1 levels precede OA development or reflect established structural and metabolic alterations. Serum WISP1 has not been extensively characterized in obese OA phenotypes, and analyses were conducted in a single cohort without external validation. OA classification was based on clinical and radiographic criteria without advanced inflammatory phenotyping, limiting characterization of disease heterogeneity. In addition, patient reported outcome measures, such as pain severity and functional status, were not assessed, which limits the ability to directly relate serum WISP1 concentrations to clinical manifestations of OA. Although multivariable models adjusted for major confounders, residual confounding from visceral adiposity, medication use, physical activity, and other metabolic factors cannot be excluded. In particular, the use of osteoarthritis-related medications, including non-steroidal anti-inflammatory

drugs, intra-articular corticosteroids, and other analgesics, was not systematically recorded or controlled for, and their potential effects on systemic inflammatory markers and circulating WISP1 levels cannot be excluded. In addition, intra-assay coefficient of variation values and extended analytical validation parameters were not assessed, which may represent a limitation.

Serum measurements do not identify the primary tissue source of WISP1; circulating levels may originate from multiple tissues, including adipose tissue, synovium, and articular cartilage, all of which have been implicated in OA and metabolic inflammation. Therefore, serum WISP1 should be interpreted as reflecting integrated systemic activity rather than tissue-specific expression. Future studies incorporating synovial fluid analysis, tissue-level expression profiling, and paired systemic local measurements are warranted to clarify the relative contribution of these sources and the absence of synovial fluid assessment limits interpretation of systemic versus intra-articular contributions. In addition, the single-center design and subgroup sample sizes may restrict generalizability. The study population was predominantly female; therefore, the findings should be interpreted in the context of a female-predominant cohort. Finally, ROC findings reflect internal discrimination and require independent validation before clinical applicability can be considered.

In conclusion, serum WISP1 concentrations were higher in obese individuals with OA compared with both obese individuals without OA and

healthy controls. This association remained statistically significant after adjustment for covariates included in the multivariable model, supporting the robustness of the observed relationship within the limits of the analysis BMI and CRP, indicating independence from these measured covariates within the present model. These findings suggest that support an association between serum WISP1 may serve as a potential marker associated with and obesity-related OA within a metabolic-inflammatory context. The observed ROC performance suggests that WISP1 may have potential for phenotypic stratification in obese populations, although its diagnostic utility cannot be inferred from cross-sectional data. Further studies are needed to clarify its biological relevance, longitudinal predictive value, and potential as a biomarker in obesity-associated OA.

Author Contributions

O.Y. conceived and designed the study, performed the statistical analyses, interpreted the data, and drafted the manuscript. O.E. contributed to study design, patient recruitment, and clinical data acquisition. S.K. performed orthopedic evaluations and radiographic assessments. C.O.I. conducted rheumatologic assessments and contributed to diagnostic confirmation of osteoarthritis cases. M.A. was responsible for laboratory analyses and biomarker measurements. S.A.Y. contributed to data management and methodological support. M.Ak. supervised the study, contributed to conceptual development. All authors contributed to drafting,

reviewing and finalizing the manuscript. All authors have read and agreed to the published version of the manuscript.

Institutional Review Board Statement

This study was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. The Local Ethics Committee of Kanuni Sultan Suleyman Training and Research Hospital approved the study (decision date: 20.02.2025, number: E-80929729-000-269138296).

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Conflicts of interest

The authors declare no conflicts of interest.

Data availability statement

The datasets used and analyzed in this study are available from the corresponding authors upon reasonable request.

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Table 1. Comparison of demographic, anthropometric and metabolic parameters among study groups.

Parameters	Healthy Control (<i>n</i> = 60)	Obese OA (-) (<i>n</i> = 60)	Obese OA (+) (<i>n</i> = 60)	<i>p</i>
Age, year	54 (48–57.5)	54 (47–58)	52.5 (48–56)	0.574 [¶]
Female sex, <i>n</i> (%)	46 (76.7)	51 (85)	55 (91.7)	0.076 [†]
Body mass index, kg/m ²	22 (21–23)	41 (39–42.2)	40 (37.3–43.5)	<0.001 ^{¶a}
Waist circumference, cm	82 (80–84)	116 (108–121)	115.7 (112–121)	<0.001 ^{¶a}
Waist-to-hip ratio	0.82 (0.81–0.83)	1.00 (0.98–1.03)	1.01 (0.96–1.05)	<0.001 ^{¶a}
Body fat percentage, (%)	23.8 (22.6–24.4)	46 (43.1–50.6)	50.6 (46.9–51.5)	<0.001 ^{¶d}
Skeletal muscle mass, kg	25.1 (24.8–25.4)	28 (25.8–31.2)	26.9 (24.9–30.9)	<0.001 ^{¶a}
Visceral fat level	6 (6–7)	21 (19–24)	22 (21–24)	<0.001 ^{¶a}
Fat-free mass index	17.5 (17.4–17.7)	20.1 (19.4–21.6)	20.3 (19.4–21.6)	<0.001 ^{¶a}

Kellgren-Lawrence grade (0/2/3/4), n (%)	-	-	13 (21.7 29.5) / 13 24 (21.7 54.5) / 24 7 (40 15.9) / 10 (16.7)	-
OA duration, year	-	-	4 (2-6.5)	-
Hypertension, n (%)	-	25 (41.7)	29 (48.3)	0.463 [†]
Dyslipidemia, n (%)	-	29 (48.3)	30 (50)	0.855 [†]
Medication, n (%)	-	44 (73.3)	42 (70)	0.685 [†]
Lipid lowering		29 (48.3)	30 (50)	0.855 [†]
ACEI/ARB		18 (30)	12 (20)	0.292 [†]
Ca ⁺⁺ channel blocker		14 (23.3)	16 (26.7)	0.833 [†]
Beta blocker		7 (11.7)	4 (6.7)	0.529 [‡]
Diuretic		4 (6.7)	18 (30)	0.002 [‡]
Leukocyte, 10 ³ /μL	7.17 ± 1.28	9.01 ± 1.57	8.62 ± 1.51	<0.001 ^{fa}
Glucose, mg/dL	86 (81-91)	102 (89-109)	98 (91-107)	<0.001 ^{¶a}
HOMA-IR	1.8 (1.3-2.1)	6.85 (3.3-11)	5 (2.5-7.99)	<0.001 ^{¶a}
Creatinine, mg/dL	0.73 ± 0.13	0.77 ± 0.14	0.77 ± 0.16	0.210 ^f
Total cholesterol, mg/dL	157 (133-179)	193.5 (168-224)	207 (183-229)	<0.001 ^{¶a}
Triglycerides, mg/dL	111.5 (77-132)	178 (144-214)	129.5 (107-182)	<0.001 ^{¶a}
HDLc, mg/dL	53 (44-59.5)	44 (39-50)	49 (41-55)	0.007 ^{¶b}
LDLc, mg/dL	112 (91-121)	129 (104-140)	131 (116-154)	<0.001 ^{¶a}
ALT, U/L	13 (11-17)	17 (13-23)	15 (12-20)	<0.001 ^{¶a}
CRP, mg/L	2.1 (1.7-2.8)	4.1 (2.6-6.4)	6.5 (4.4-7.3)	<0.001 ^{¶d}
ESR, mm/h	11 (6.5-15.5)	11 (6-15)	16.5 (13-22)	<0.001 ^{¶c}

OA, osteoarthritis; ACEI/ARB, angiotensin-converting enzyme inhibitor/angiotensin II receptor blocker; HOMA-IR, homeostatic model assessment of insulin resistance; HDLc, high-density lipoprotein cholesterol; LDLc, low-density lipoprotein cholesterol; ALT, alanine aminotransferase; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate. Data are presented as mean ± standard deviation (SD), median (interquartile range, Q1-Q3), or number (percentage), as appropriate. [†]Pearson's chi-square test; ^f one-way analysis of variance (ANOVA); [¶]Kruskal-Wallis H test. Pairwise comparisons were performed using the Mann-Whitney U test following a significant Kruskal-Wallis test and Bonferroni-adjusted post hoc analysis following one-way ANOVA. a: HC < Obese OA (-) and Obese OA (+); b: Obese OA (-) < HC; c: Obese OA (+) > Obese OA (-) and HC; d: Obese OA (+) > Obese OA (-) > HC. A p value < 0.05 was considered statistically significant.

Table 2. Comparison of serum WISP1 levels among healthy controls, knee osteoarthritis, and non-knee osteoarthritis groups.

	Healthy Control (n = 60)	Knee OA (+) (n = 44)	Non-knee OA (+) (n = 16)	p
WISP1, pg/mL	258.5 (211.5-285.5)	506 (468.5-1151)	554 (475.5-1106)	<0.001 ^{¶a}

Data are presented as median (interquartile range, Q1-Q3). [¶] Kruskal-Wallis H test. Pairwise comparisons were performed using the Mann-Whitney U test following a significant Kruskal-

Wallis test. a: HC < Knee OA (+) and Non-knee OA (+). A p value < 0.05 was considered statistically significant.

Table 3. Univariate and multivariable binary logistic regression analysis of factors associated with osteoarthritis in obese adults.

Variable	Univariate Model				Multivariable Model			
	β	p	OR	95% CI	β	p	OR	95% CI
Age	-0.09	0.656	0.991	0.951-1.032	0.0142	0.57	1.014	0.967-1.063
Gender	0.663	0.261	1.941	0.610-6.178	0.7676	0.28	2.154	0.525-8.835
BMI	-0.26	0.615	0.974	0.881-1.078	0.0152	0.81	1.015	0.897-1.149
WISP1	0.003	0.001	1.003	1.001-1.004	0.00301	<0.0	1.003	1.001-1.005
CRP	0.062	0.289	1.064	0.949-1.193	0.0802	0.23	1.083	0.950-1.236
HOMA-IR	-	0.118	-	-	-	-	-	-
Diuretic use	0.0581.792	0.002	0.943	0.877-1.015	0.1111.523	0.020	0.895	0.815-0.983
			6.000	1.890-19.043		0.018	4.585	1.299-16.186

β , regression coefficient; OR, odds ratio; CI, confidence interval; BMI, body mass index; CRP, C-reactive protein; HOMA-IR, homeostasis model assessment of insulin resistance. Binary logistic regression analysis was performed. In the univariate analyses, separate regression models were constructed for each independent variable. In the multivariable analysis, variables were entered simultaneously into a single model. Model fit was assessed using the Hosmer-Lemeshow goodness-of-fit test, and the model demonstrated an adequate fit ($p = 0.314$). A p value < 0.05 was considered statistically significant.

Table 4: Multinomial logistic regression analysis comparing obese OA (-) and obese OA (+) groups with healthy controls.

Variable	Obese OA (-) vs HC				Obese OA (+) vs HC			
	β	p	OR	95% CI	β	p	OR	95% CI
Age	0.014	0.787	1.015	0.914-1.127	0.026	0.639	1.026	0.922-1.142
WISP1	0.025	<0.001	1.025	1.011-1.039	0.028	<0.001	1.028	1.014-1.042
HOMA-IR	2.477	0.011	11.902	1.775-79.801	2.357	0.015	10.563	1.575-70.837
CRP	1.367	0.002	3.925	1.642-9.382	1.455	0.001	4.284	1.788-10.263

OA, osteoarthritis; HC, healthy control; β , regression coefficient; CI, confidence interval; BMI, body mass index; CRP, C-reactive protein. In the multinomial logistic regression analysis, the healthy control group was used as the reference category. Results are

presented as β coefficients, odds ratios (OR) values, and 95% confidence intervals (CI). A p value < 0.05 was considered statistically significant.

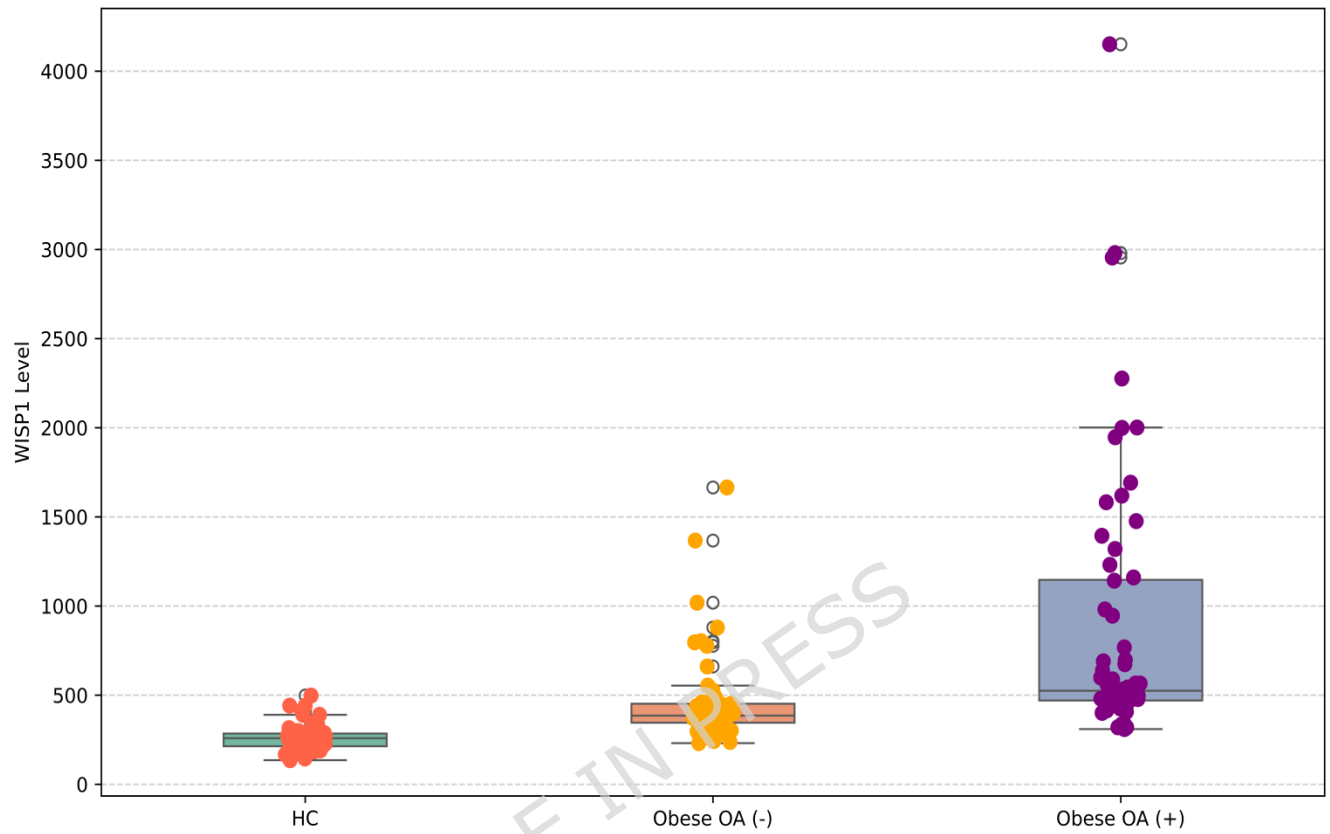


Figure 1. Distribution of serum WISP1 levels among healthy controls, obese individuals without osteoarthritis, and obese individuals with osteoarthritis. The box plots represent the median and interquartile range (IQR), and individual measurements are presented as jittered scatter points.

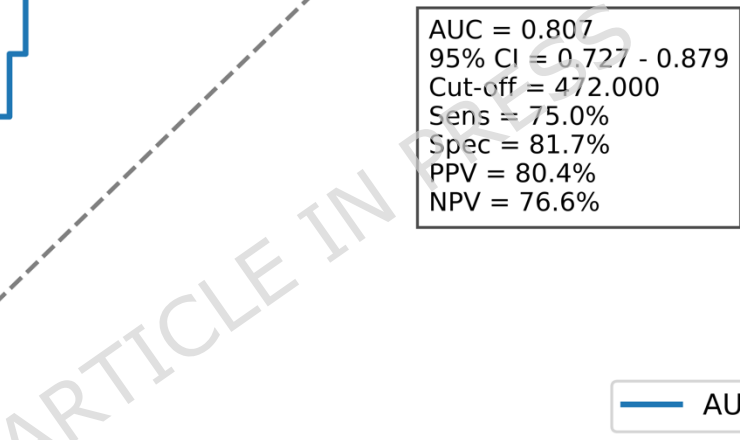


Table captions and figure legends

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Table 4: Multinomial logistic regression analysis comparing obese OA (-) and obese OA (+) groups with healthy controls.

Figure 1. Distribution of serum WISP1 levels among healthy controls, obese individuals without osteoarthritis, and obese individuals with osteoarthritis. The box plots represent the median and interquartile range (IQR), and individual measurements are presented as jittered scatter points.

Figure 2 3. Receiver operating characteristic curve demonstrating the discriminative performance of serum WISP1 levels in distinguishing obese individuals without osteoarthritis from obese individuals with osteoarthritis. The area under the curve (AUC) was 0.807, with a 95% confidence interval of 0.727–0.879. TPR, true positive rate; FPR, false positive rate; PPV, positive predictive value; NPV, negative predictive value.